

Provenance

Multi-Modal Video Forensics and Integrity System

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1 Purpose of This Document

This document defines the functional, technical, and architectural requirements for Provenance, a video forensics system. It serves as a single source of truth for reviewers, faculty, and technical evaluators to understand the system’s scope, design decisions, machine learning methodology, and operational constraints.

2 Background and Motivation

Video has become the most trusted and persuasive form of digital evidence. With the rise of high-quality generative models and widespread content resharing, video is increasingly exploited to mislead audiences through:

- Facial manipulation and identity alteration
- Fully AI-generated synthetic videos
- Misrepresentation of real footage through false claims about time, place, or event

Human visual inspection is no longer sufficient to reliably assess such content, particularly under time pressure. Existing verification workflows are fragmented, reactive, and often opaque, creating a gap between content dissemination and credibility assessment.

3 Problem Statement

3.1 Core Problem

There is no unified, explainable system that assists users in assessing the credibility risk of video content across multiple manipulation dimensions at the time the content is being shared.

3.2 Key Challenges

- Modern manipulations are visually convincing and non-obvious
- AI-generated videos lack traditional camera artifacts
- Contextual misuse exploits trust in "real" footage
- Existing tools address only isolated aspects of the problem

4 Product Vision

Provenance is an assistive forensic system that evaluates video content across three independent yet complementary analytical dimensions, producing risk-based, explainable outputs rather than definitive judgments.

5 Scope Definition

In Scope

- Short-form video analysis (5 to 20 seconds)
- Multi-modal forensic signals
- Explainable risk scoring
- CPU inference with optional GPU acceleration

- Modular, extensible architecture

Out of Scope

- Truth determination
- Identity attribution
- Content enforcement
- Long-form live-stream ingestion (future extension)

6 High-Level System Capabilities

Provenance performs three primary analyses, each executable independently on the same video input:

1. Deepfake Risk Analysis
2. AI-Generated Video Detection
3. Context Integrity (Miscontext) Analysis

7 User Personas

7.1 Journalist / Media Verifier

Needs rapid credibility signals, operates under time pressure, and requires interpretability.

7.2 Fact-Checking Analyst

Performs deeper verification and needs evidence-backed signals.

7.3 Platform or Research Analyst

Reviews high volumes of content and uses risk scores for prioritization.

8 User Workflow

1. User uploads a short video clip
2. User optionally provides an associated claim
3. User selects one or more analysis modes
4. System processes the video and generates forensic signals
5. Results are presented with explanations and limitations

9 Functional Requirements

- **FR-1 Video Ingestion:** accept MP4 and MOV formats; enforce duration and size constraints; validate readability.
- **FR-2 Frame Extraction:** temporal sampling at fixed intervals with preprocessing suitable for each model.
- **FR-3 Independent Analysis Execution:** each analysis module operates in isolation; a failure in one module does not block others.

10 Deepfake Risk Analysis Module

10.1 Objective

Assess whether a video contains facial manipulations or deepfake artifacts indicative of synthetic alteration.

10.2 Model Strategy

- Face detection with OpenCV YuNet on each sampled frame
- A pretrained Vision Transformer face-forgery classifier scored on each detected face
- Aggregation of per-face scores into a manipulation-likelihood risk score

10.3 Model Basis

The classifier is pretrained on real and fake face corpora derived from FaceForensics-style datasets and modern face-swapping and reenactment methods. No fine-tuning is performed on user data. Videos containing no faces return a LOW result, since there is no face to assess.

10.4 Inference Pipeline

1. Frame extraction
2. Face detection and cropping
3. Per-face classification
4. Aggregation across faces
5. Confidence and consistency reporting

10.5 Output Signals

- Deepfake manipulation likelihood
- Face-level detection consistency

11 AI-Generated Video Detection Module

11.1 Objective

Detect whether a video is likely synthetically generated rather than camera-captured.

11.2 Model Strategy

- A pretrained real-versus-artificial image classifier (Vision Transformer)
- Per-frame classification across the sampled frames
- Aggregation of the mean artificial probability and per-frame agreement

11.3 Model Basis

The detector uses a pretrained classifier calibrated on a mixture of real camera footage and AI-generated imagery from diffusion and transformer-based generators. Whole-frame analysis lets it flag fully generated content, not just faces.

11.4 Output Signals

- Synthetic media likelihood
- Temporal feature coherence
- Per-frame synthetic ratio

12 Context Integrity (Miscontext) Analysis Module

12.1 Objective

Identify whether real video footage is being misrepresented via misleading claims.

12.2 Signals

- Temporal reuse detection
- Scene-to-claim semantic alignment
- Environmental consistency (lighting and time-of-day cues)

12.3 Methodology

- CLIP zero-shot scene classification, comparing the dominant detected scene against the scene implied by the claim
- ResNet-50 frame-embedding similarity against a reference set for reuse detection
- Brightness-based lighting classification checked against time cues parsed from the claim

13 Risk Scoring Engine

13.1 Scoring Philosophy

The system produces risk indicators, not truth labels.

13.2 Aggregation

Weighted signal combination normalized to a 0 to 100 scale.

13.3 Risk Bands

LOW, MEDIUM, and HIGH.

14 Explainability Layer

Each result includes signal-level outcomes, human-readable explanations, an executive summary, and explicit uncertainty disclosure.

15 System Architecture Overview

Web interface (React, Vite) to API gateway (FastAPI) to a video processing layer, then the three independent analysis modules, a risk scoring engine, and an explainable output renderer.

16 Performance Requirements

- Per-module analysis of roughly 3 to 15 seconds per clip on CPU
- GPU acceleration with automatic CPU fallback
- Stable behavior under repeated use

17 Security and Privacy

- Stateless processing with no permanent storage
- No user accounts or stored credentials
- No biometric identification
- Hosted deployments served over HTTPS

18 Failure Modes and Mitigation

- **False positives:** mitigated with confidence bands and detailed explanations.
- **Low-quality video:** mitigated through signal weighting and graceful degradation.
- **Partial module failure:** mitigated by independent module execution.

19 Ethical Considerations

- No automated enforcement
- Transparency-first design
- Human judgment remains the final authority

20 Deployment Strategy

- Containerized backend (Docker), deployable on Hugging Face Spaces
- Static frontend deployable on Vercel, configured with the backend URL
- Modular scaling per analysis module

21 Validation and Acceptance Criteria

- All three analyses operate independently
- Outputs are interpretable
- The system remains stable under repeated use
- Results align with documented limitations

22 Future Extensions

- Audio-based manipulation analysis
- Live-stream sampling
- Cross-video coordination detection
- Browser-level integrations

23 Assumptions and Constraints

- Short-form videos only
- Limited reference corpus
- Near-live analysis focus

24 Final Summary

Provenance is a multi-modal, research-informed video forensics system that integrates deepfake detection, synthetic media analysis, and contextual verification into a single, explainable platform. By prioritizing interpretability, modularity, and responsible deployment, it demonstrates how machine learning can be applied to real-world credibility assessment without overclaiming or ethical compromise.